In [1]:
```
import akshare as ak
from __future__ import (absolute_import, division, print_function, unicode_literals)
import datetime
import backtrader as bt
import backtrader.feeds as btfeeds
import backtrader.indicators as btind
import pandas as pd
import numpy as np
```

1.获取数据

In [3]:
```
df = ak.futures_zh_minute_sina(symbol="FU0", period="1")
print(df)
df.index=pd.to_datetime(df.datetime)
df=df[['open','high','low','close','volume']]
df.to_csv('fuo.csv' )
```

```
                 datetime    open    high     low   close  volume    hold
0     2022-12-06 21:13:00  2526.0  2532.0  2525.0  2529.0    6767  373941
1     2022-12-06 21:14:00  2530.0  2530.0  2527.0  2529.0    3071  373906
2     2022-12-06 21:15:00  2529.0  2533.0  2528.0  2532.0    4242  373586
3     2022-12-06 21:16:00  2530.0  2534.0  2530.0  2533.0    5002  373267
4     2022-12-06 21:17:00  2532.0  2533.0  2530.0  2531.0    2596  373233
...                   ...     ...     ...     ...     ...     ...     ...
1018  2022-12-09 14:56:00  2541.0  2542.0  2539.0  2541.0    2773  391391
1019  2022-12-09 14:57:00  2542.0  2542.0  2539.0  2541.0    2356  391088
1020  2022-12-09 14:58:00  2541.0  2544.0  2540.0  2543.0    2346  390558
1021  2022-12-09 14:59:00  2543.0  2546.0  2543.0  2546.0    4575  389586
1022  2022-12-09 15:00:00  2545.0  2561.0  2545.0  2561.0    1539  385696

[1023 rows x 7 columns]
```

2.自定义指标

In [2]:
```
class DT_Line(bt.Indicator):
    lines = ('U', 'D')
    params = (('period', 2), ('k_u', 0.7), ('k_d', 0.7))


    def __init__(self):
        self.addminperiod(self.p.period + 1)


    def next(self):
        HH = max(self.data.high.get(-1, size=self.p.period))
        LC = min(self.data.close.get(-1, size=self.p.period))
        HC = max(self.data.close.get(-1, size=self.p.period))
        LL = min(self.data.low.get(-1, size=self.p.period))
        R = max(HH - LC, HC - LL)
        self.lines.U[0] = self.data.open[0] + self.p.k_u * R
        self.lines.D[0] = self.data.open[0] - self.p.k_d * R
```

3.核心策略

In [3]:
```
class DualThrust(bt.Strategy):

    def __init__(self):
        self.dataclose = self.data0.close
```

```python
        self.D_Line = DT_Line(self.data1)
        self.D_Line = self.D_Line()
        self.D_Line.plotinfo.plotmaster = self.data0


        self.buy_signal = bt.indicators.CrossOver(self.dataclose, self.D_Line.U)
        self.sell_signal = bt.indicators.CrossDown(self.dataclose, self.D_Line.D)

    def start(self):
        print("开始赚钱！")

    def prenext(self):
        print("not mature")


    def next(self):
        if self.data.datetime.time() > datetime.time(9, 3) and self.data.datetime.ti
            if not self.position and self.buy_signal[0] == 1:
                self.order = self.buy()
            if not self.position and self.sell_signal[0] == 1:
                self.order = self.sell()

            if self.getposition().size < 0 and self.buy_signal[0] == 1:
                self.order = self.close()
                self.order = self.buy()

            if self.getposition().size > 0 and self.sell_signal[0] == 1:
                self.order = self.close()
                self.order = self.sell()

        if self.data.datetime.time() >= datetime.time(22, 55) and self.position:
            self.order = self.close()


    def stop(self):
```

4.回测（短数据）

In [7]:

```python
if __name__ == '__main__':
    # 1. Create a cerebro
    cerebro = bt.Cerebro()
    # 2. Add data feed
    # 2.1 Creat a data feed
    dataframe = pd.read_csv('fuo.csv')
    dataframe['datetime'] = pd.to_datetime(dataframe['datetime'])
    dataframe.set_index('datetime', inplace=True)
    brf_min_bar = bt.feeds.PandasData(dataname=dataframe,
                                      fromdate=datetime.datetime(2022, 12, 6),
                                      todate=datetime.datetime(2022, 12, 9),
                                      timeframe=bt.TimeFrame.Minutes
                                      )
    # 2.2Add the Data Feed to Cerebro
    cerebro.adddata(brf_min_bar)
    cerebro.resampledata(brf_min_bar, timeframe=bt.TimeFrame.Days)

    # 3 Add strategy
    cerebro.addstrategy(DualThrust)
    # 4. Run
    cerebro.run()
    # 5.Plot result
    cerebro.plot(style='candle')
```

开始赚钱！
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature

not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature

not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature

not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature

not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature

not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature

not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature

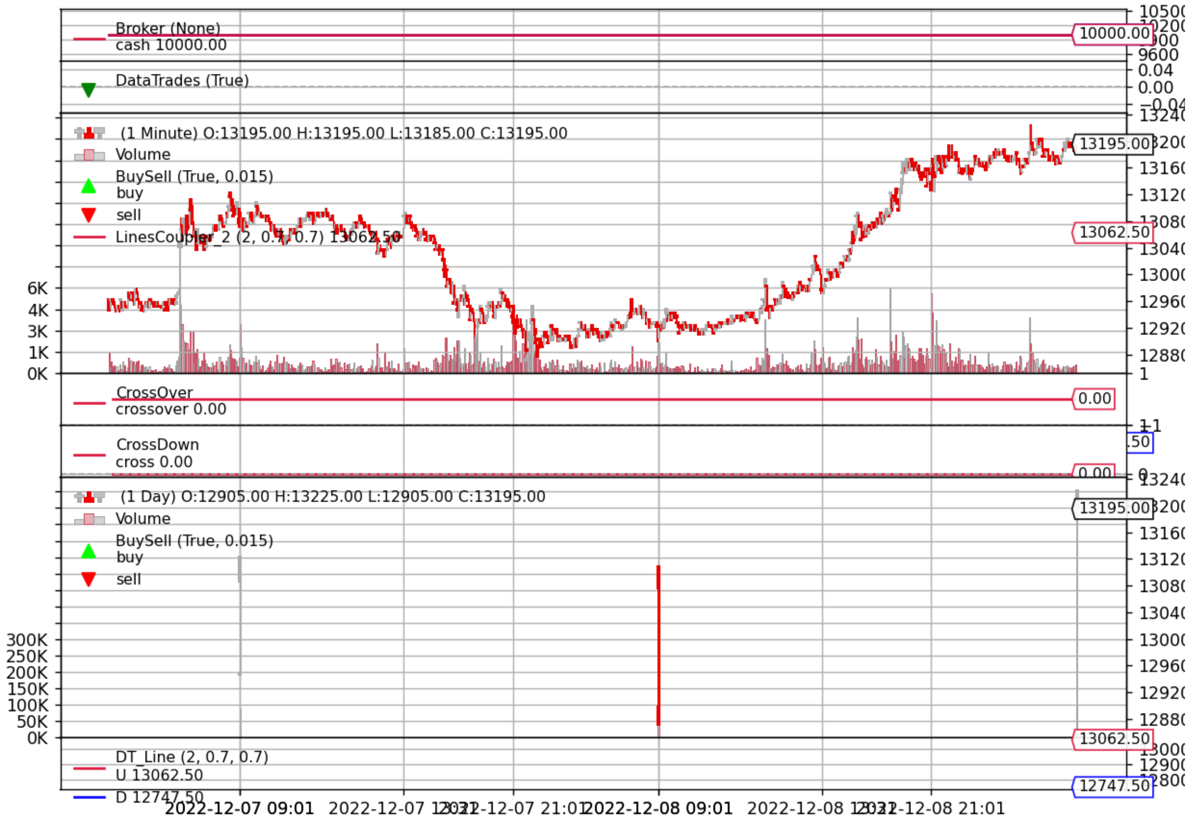
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature

not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature

not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature

not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature

not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
over



5.回测（长数据）

In [8]:

```
if __name__ == '__main__':
    # 1. Create a cerebro
    cerebro = bt.Cerebro()
    # 2. Add data feed
    # 2.1 Creat a data feed
    dataframe = pd.read_csv('rbfi_min.csv')
    dataframe['datetime'] = pd.to_datetime(dataframe['datetime'])
    dataframe.set_index('datetime', inplace=True)
    brf_min_bar = bt.feeds.PandasData(dataname=dataframe,
                                      fromdate=datetime.datetime(2018, 1, 26),
                                      todate=datetime.datetime(2018, 6, 22),
                                      timeframe=bt.TimeFrame.Minutes
                                      )
    # 2.2Add the Data Feed to Cerebro
    cerebro.adddata(brf_min_bar)
    cerebro.resampledata(brf_min_bar, timeframe=bt.TimeFrame.Days)

    # 3 Add strategy
    cerebro.addstrategy(DualThrust)
    # 4. Run
    cerebro.run()
    # 5.Plot result
    cerebro.plot(style='candle')
```

```
开始赚钱！
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
```

not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature

not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature

not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature

not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature

not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature

not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature

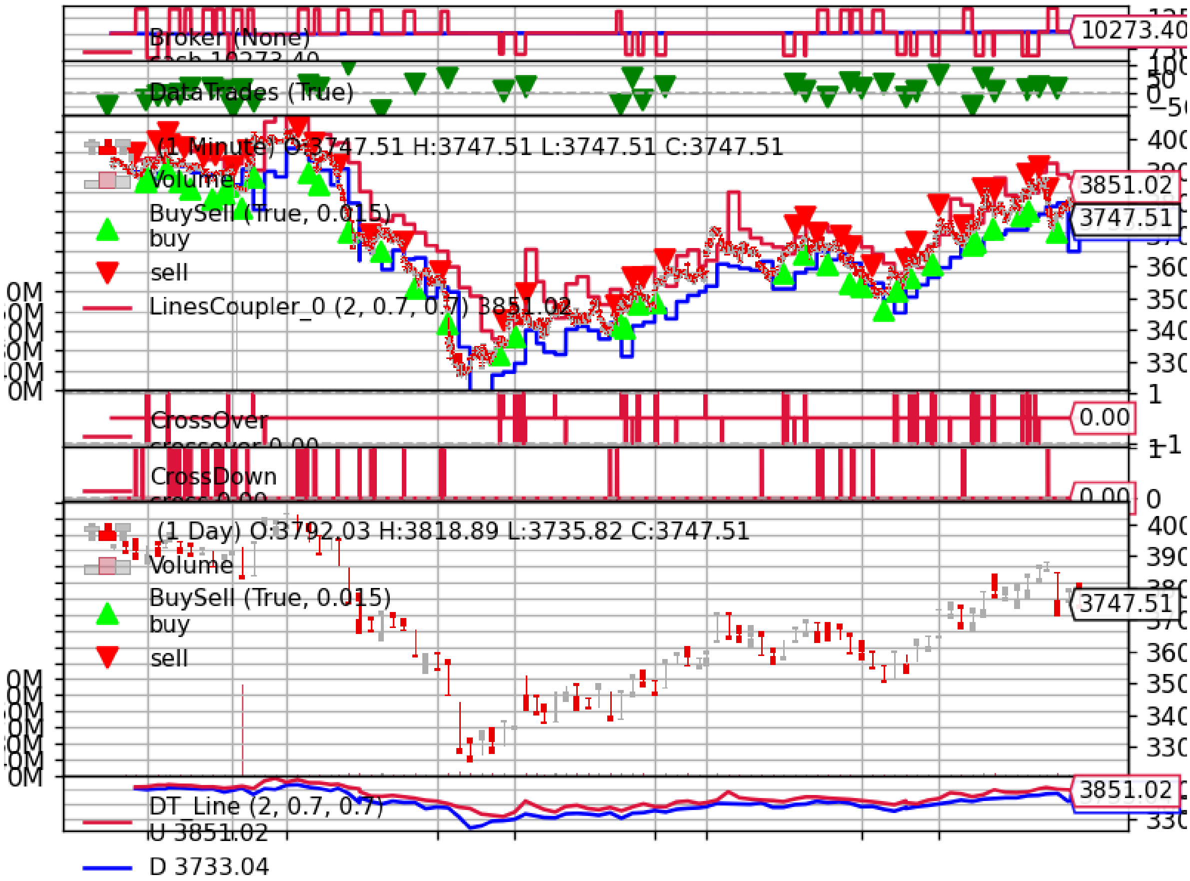
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature

not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature

not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature

not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature

not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
over



6.图像调试

In [6]:

```python
if __name__ == '__main__':
    cerebro = bt.Cerebro(stdstats=False)
    # cerebro.addobserver(bt.observers.Broker)
    cerebro.addobserver(bt.observers.Trades)
    cerebro.addobserver(bt.observers.BuySell)
    cerebro.addobserver(bt.observers.DrawDown)
    cerebro.addobserver(bt.observers.Value)
    cerebro.addobserver(bt.observers.TimeReturn)


    # 2. Add data feed
    # 2.1 Creat a data feed
    dataframe = pd.read_csv('rbfi_min.csv')
    dataframe['datetime'] = pd.to_datetime(dataframe['datetime'])
    dataframe.set_index('datetime', inplace=True)
    brf_min_bar = bt.feeds.PandasData(dataname=dataframe,
                                      fromdate=datetime.datetime(2018, 1, 26),
                                      todate=datetime.datetime(2018, 6, 22),
                                      timeframe=bt.TimeFrame.Minutes
                                      )
    # 2.2Add the Data Feed to Cerebro
    cerebro.adddata(brf_min_bar)
    cerebro.resampledata(brf_min_bar, timeframe=bt.TimeFrame.Days)

    # 3 Add strategy
    cerebro.addstrategy(DualThrust)


    # 4. Run
    cerebro.run()
    # 5.Plot result
    cerebro.plot(style='candle')
```

```
开始赚钱!
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
```

not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature

not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature

not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature

not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature

not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature

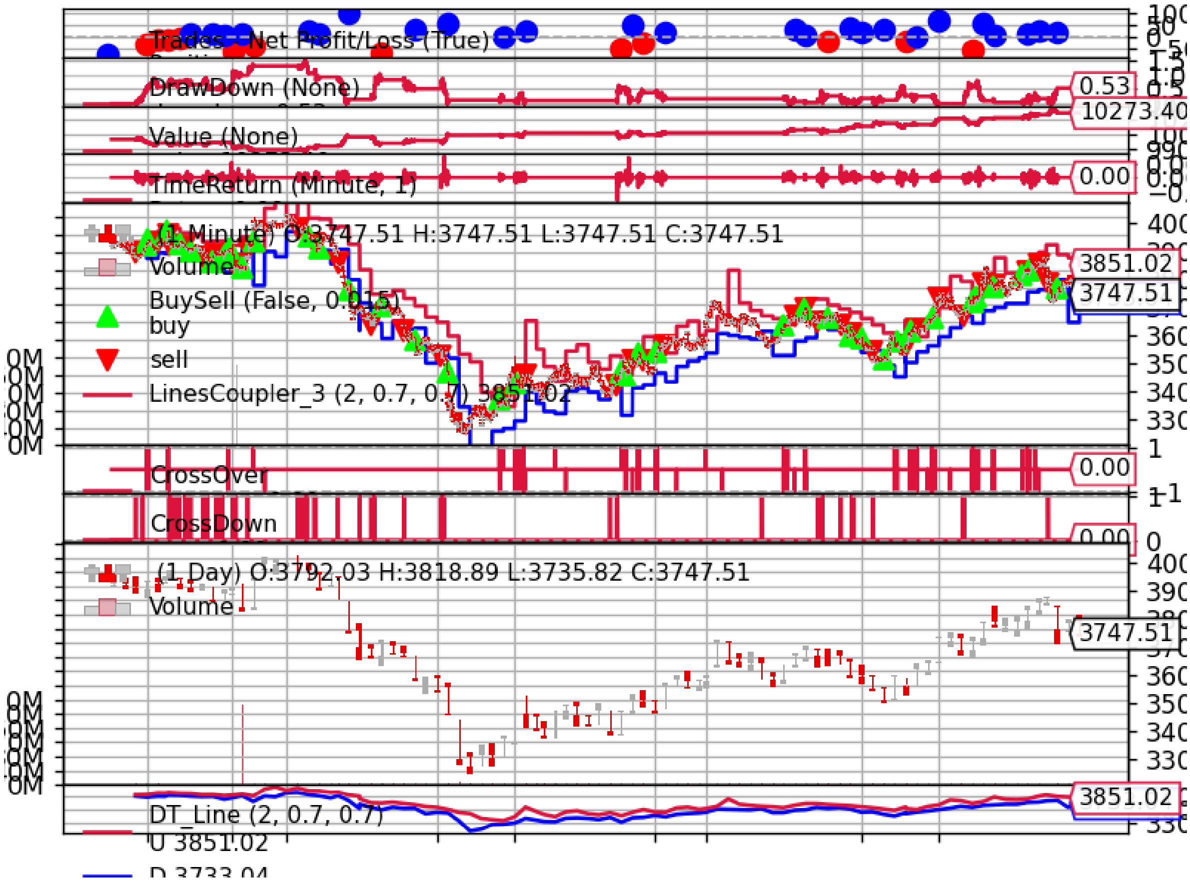
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature

not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature

not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature

not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature

not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
over



In [7]:

```python
if __name__ == '__main__':
    cerebro = bt.Cerebro(stdstats=False)
    # cerebro.addobserver(bt.observers.Broker)
    cerebro.addobserver(bt.observers.Trades)
    cerebro.addobserver(bt.observers.BuySell)
    cerebro.addobserver(bt.observers.DrawDown)
    cerebro.addobserver(bt.observers.Value)
    cerebro.addobserver(bt.observers.TimeReturn)

    cerebro.addanalyzer(bt.analyzers.DrawDown)
    cerebro.addanalyzer(bt.analyzers.TradeAnalyzer)

    # 2. Add data feed
    # 2.1 Creat a data feed
    dataframe = pd.read_csv('rbfi_min.csv')
    dataframe['datetime'] = pd.to_datetime(dataframe['datetime'])
    dataframe.set_index('datetime', inplace=True)
    brf_min_bar = bt.feeds.PandasData(dataname=dataframe,
                                      fromdate=datetime.datetime(2018, 1, 26),
                                      todate=datetime.datetime(2018, 6, 22),
                                      timeframe=bt.TimeFrame.Minutes
                                      )
    # 2.2Add the Data Feed to Cerebro
    cerebro.adddata(brf_min_bar)
    cerebro.resampledata(brf_min_bar, timeframe=bt.TimeFrame.Days)

    # 3 Add strategy
    cerebro.addstrategy(DualThrust)


    # 4. Run
    res = cerebro.run()[0]
    drow_down_data = res.analyzers.drawdown.get_analysis()
    print('==========Drow Down=============')
    print('max drowdown:%s %%' % drow_down_data['max']['drawdown'])
```

```python
    print('max money drowdown:%s' % drow_down_data['max']['moneydown'])
```

```
开始赚钱！
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
```

not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature

not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature

not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature

not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature

not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature

not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature

not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature

not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature

not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature

not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature
not mature

```
over
==========Drow Down==============
max drowdown:1.5285187421710018 %
max money drowdown:153.13999999999942
```